

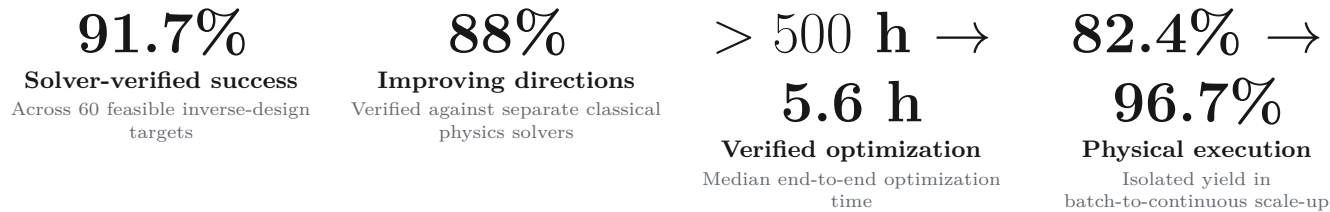
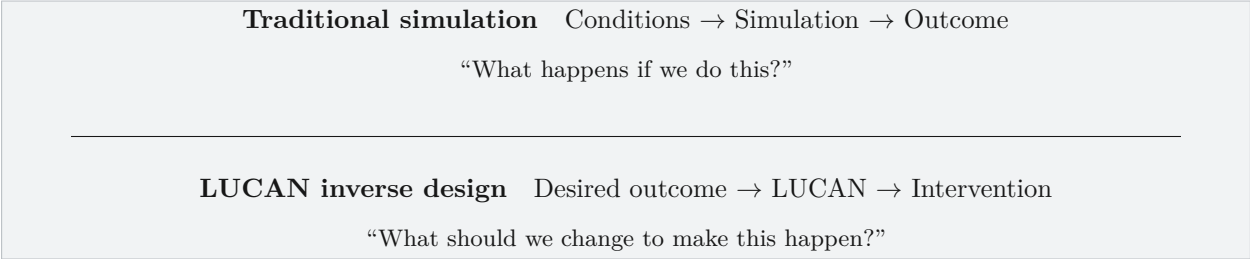
LUCAN

A Foundation World Model for Physical Intelligence

Designing physical systems from molecules to manufacturing through cross-scale inverse design.

One model. Multiple physical scales. One design objective.

LUCAN is designed to connect molecular, thermodynamic, fluid, biological, and process-scale representations within a shared differentiable computational system—so the model can move beyond predicting outcomes toward proposing interventions that achieve them.



LUCAN in 60 seconds

The physical world is not organized into software modules. Manufacturing outcomes emerge from interactions across scales.

The problem	Molecular simulation, thermodynamics, CFD, biological mechanics, and process control are typically separated into different tools. Engineers connect them through hand-offs and repeated forward simulation.
What LUCAN does	LUCAN maps different scientific representations into a shared Sparse Mixture-of-Experts architecture so cross-scale relationships can be learned inside one differentiable computational system.
Why it matters	A differentiable cross-scale representation can provide useful directions through the design space. Instead of only predicting the consequence of a proposed process, the model can help identify molecular and process variables that should change to reach a target outcome.
How we test it	We test increasingly demanding cross-scale behaviors, verify inverse-design proposals against independent classical numerical solvers, and then evaluate selected interventions through prospective physical execution.

The evidence ladder

MODEL TEST A frozen model is evaluated on predefined held-out computational tasks.

PHYSICS VERIFIED A model-generated intervention is independently checked using high-fidelity classical numerical physics.

PHYSICALLY EXECUTED A frozen model-generated intervention is carried into a laboratory, pilot, or industrial environment.

What this whitepaper demonstrates

1. A shared architecture can route and combine information across molecular, continuum, and biological physical representations.
2. Controlled changes can propagate upward from microscopic energetics to reactor-scale thermodynamics and downward from reactor hydrodynamics to microscopic structural response.
3. The model can represent nonlinear trade-offs where improving one mechanism worsens another.
4. Cross-scale gradients can produce interventions that improve independently evaluated physical objectives.
5. Selected model-generated operating regimes can transfer from computation into prospective industrial execution.

The central idea: LUCAN's goal is to move scientific AI from predicting physical systems toward **designing the interventions required to achieve targeted physical outcomes**.

One physical system, multiple interacting scales

LUCAN is designed to reason across the connected physical hierarchy rather than treating molecular behavior, process physics, and equipment-scale outcomes as isolated problems.



Conceptual illustration. LUCAN connects molecular-scale variables, process-scale physics, and equipment-scale outcomes inside a shared computational design space.

From molecules to manufacturing: the objective is not simply to predict each scale independently, but to understand how changes at one level influence outcomes at another—and use those relationships to design better interventions.

1 Why Physical AI Breaks at Scale

Physical products are created across many scales at once. Molecular energetics influence reaction rates and heat release; reactor geometry and fluid flow determine transport; local shear and mass transfer affect biological or material response; and all of these interactions shape final manufacturing yield, quality, and cost.

Most computational workflows model these layers separately. A molecule may be optimized in one environment, kinetics in another, fluid dynamics in a third, and process controls in a fourth. Engineers then connect the outputs manually and iterate forward.

That workflow is good at answering:

“If we choose these conditions, what happens?”

The harder industrial question is the inverse:

“Given the outcome we want, what should we change?”

1.1 What LUCAN Is

LUCAN is a physical world model designed to connect decisions at the molecular level to outcomes at manufacturing scale within one differentiable computational system.

It is not a video world model and it is not a single-domain surrogate. LUCAN operates on typed scientific representations—including molecular graphs, three-dimensional physical fields, and deforming geometries—and maps them into a shared Sparse Mixture-of-Experts (MoE) architecture.

The objective is to preserve enough of the relationship between scales that information can move in both directions: from microscopic interventions to macroscopic consequences, and from factory-scale objectives back toward molecular and process variables that can be changed.

1.2 Why Differentiability Changes the Workflow

A conventional engineering loop typically looks like:

propose → simulate → evaluate → adjust → repeat.

LUCAN is designed to support a different loop:

specify outcome → generate intervention → verify.

In plain language, differentiability gives the model a way to estimate not only what outcome follows from a design, but also **which controllable variables should move—and in which direction—to improve the desired outcome**. We call this capability **Cross-Scale Inverse Design**.

The remainder of this whitepaper tests the pieces required for that claim: shared computation across domains, micro-to-macro propagation, macro-to-micro feedback, nonlinear multiphysics trade-offs, full-chain integration, independent solver verification, and prospective physical execution.

2 How LUCAN Is Tested

As a commercial entity building proprietary industrial infrastructure, Shodh AI does not disclose model weights, training corpus composition, or internal architectural implementations. This technical report therefore functions as a **capability release**. Our objective is to provide rigorous evidence of cross-scale physical intelligence through frozen data holdouts, controlled cross-scale interventions, model-independent classical verification, predefined pass/fail criteria, and prospective physical execution.

2.1 Unified Sparse-MoE Architecture

The foundation model evaluated in this report is **LUCAN**, a Sparse Mixture-of-Experts (MoE) neural architecture.

To evaluate coupled physical phenomena without catastrophic loss of geometric or topological fidelity, the model does not collapse fundamentally different physical domains into undifferentiated patches. Instead, it preserves domain-specific scientific representations—including discrete molecular graphs, dense three-dimensional space-time fields, and deforming irregular meshes—which are projected into a shared continuous latent interface.

This design allows the MoE routing layer to dynamically allocate expert computation across molecular, mesoscale, and continuum representations within a unified computational framework.

Crucially, the architecture provides a **shared, end-to-end differentiable computational path across physical scales**—the mathematical prerequisite for the cross-scale inverse-design experiments evaluated in Section 4.

2.2 Fail-Closed Falsification and Leakage Controls

In physical machine learning, neural surrogates can achieve artificially low error rates through train-test leakage, interpolation between related simulation states, or memorization of known geometries and molecular families.

To ensure that the capabilities evaluated in this report cannot be explained by direct data overlap or trivial interpolation within known parent simulations, we subject the model to a strict **Fail-Closed falsification protocol**.

A capability claim is rejected from evaluation unless its underlying test data satisfies the following controls:

- **Temporal-Parent Holdouts:** All temporal windows originating from the same parent simulation trajectory are assigned entirely to either training or testing, ensuring zero temporal overlap. The model therefore cannot obtain an apparent generalization advantage by interpolating between neighboring states from a known trajectory.
- **Topological and Scaffold Holdouts:** Spatial meshes are grouped according to their underlying base CAD geometry. Changes in mesh resolution, orientation, or impeller rotation are not treated as novel geometries. Molecular datasets are partitioned using deterministic structural frameworks, including

Murcko scaffolds, with entire reaction templates and metal-ligand motifs withheld from the training corpus where applicable.

- **Pretraining Contamination Audits:** Test indices are verified against the foundational pretraining corpus using SHA-256 hashing together with rule-defined approximate-membership queries to detect and exclude potential contamination.

These controls are designed to remove trivial explanations for successful prediction before any claim of cross-scale capability is considered.

2.3 Model-Independent High-Fidelity Numerical Verification

The foundation model’s neural predictions are never treated as physical ground truth.

Forward predictions, cross-scale interventions, and inverse-design proposals are instead judged against established high-fidelity classical numerical solvers operating under matched physical conditions.

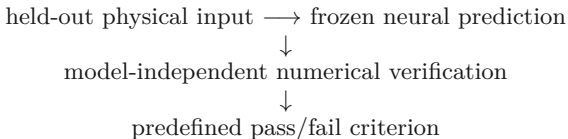
At the microscopic scale, molecular and transition-state geometries are verified using Density Functional Theory (DFT) calculations performed with ORCA v5.0.4 [5].

At the macroscopic scale, continuous-flow, transport, and moving-boundary physics are evaluated against FP64 OpenFOAM v2312 numerical trajectories [6].

These classical solvers therefore function as **model-independent numerical arbiters**: the neural model generates a prediction or design direction, while the relevant conventional physics solver determines whether that prediction remains physically valid under the corresponding governing equations and numerical constraints.

For each experiment, the appropriate physical quantities, field errors, conservation constraints, or feasibility conditions are evaluated against predefined acceptance criteria before a capability is classified as validated.

Accordingly, every capability result reported in the following sections is evaluated under the same hierarchy:



Where prospective physical execution is available, numerical verification is followed by evaluation in the corresponding real-world industrial environment.

3 Evidence of Cross-Scale Physical Reasoning

To validate the hypothesis of a unified World Model, we must establish that the architecture has learned a shared physical representation capable of propagating physically meaningful interventional relationships across scales that are traditionally modeled and optimized separately.

We evaluate this capability through five physical tests of increasing difficulty: (1) **shared domain-responsive**

computation, (2) **Micro-to-Macro controlled intervention**, (3) **Macro-to-Micro multiphysics feedback**, (4) **nonlinear competing-mechanism trade-offs**, and (5) **end-to-end molecule-to-reactor-to-cell integration**.

3.1 Test 1: Does One Model Share Computation Across Physical Domains?

A physical foundation model must process highly divergent domains—such as discrete molecular graphs and continuous turbulent fields—through substantially overlapping computational pathways. If the architecture rigidly partitions computation by scale, it functions primarily as a collection of independently routed domain models rather than a unified World Model capable of cross-domain knowledge transfer.

Objective & Setup: We evaluated a frozen checkpoint to determine whether a single neural engine could concurrently process microscopic molecular representations and macroscopic reactor-scale fluid-dynamics representations.

Empirical Results: LUCAN demonstrated substantial shared computation alongside statistically significant domain specialization, without suffering from global expert collapse.

- **Network-Wide Shared Overlap:** The mean cross-domain expert pathway overlap was **52%**. This establishes substantial shared computation between microscopic molecular and macroscopic reactor representations.
- **Domain-Responsive Routing:** Cross-domain routing divergence was $45\times$ **greater** than the corresponding within-domain routing noise, showing that the shared backbone responds strongly to physical modality.
- **Statistical Significance:** Permutation testing yielded $p = 0.00004$, rejecting the hypothesis that the observed routing separation arose from stochastic assignment noise.
- **Result: PASS.** The frozen checkpoint retains substantial shared computation between molecular and reactor representations while exhibiting strong domain-responsive routing.

Together, these results establish the representational prerequisite for cross-scale reasoning: a shared computational substrate that remains responsive to the distinct structure of different physical domains.

3.2 Test 2: Can a Molecular Change Propagate to Reactor Scale?

A generalized World Model must not only process multiple physical scales concurrently; it must also preserve physically meaningful relationships between them. In an industrial reacting system, microscopic molecular energetics directly influence macroscopic heat generation, temperature evolution, and fluid behavior.

We therefore evaluated whether an isolated intervention on a microscopic energetic variable could propagate

Routing audit	Observed result	Statistical test	Verdict
Mean cross-domain pathway overlap	52%	—	PASS
Cross-domain divergence / within-domain noise	45×	$p = 0.00004$	PASS

Figure 1. Frozen-checkpoint routing audit. Molecular and reactor representations retain 52% mean pathway overlap, while their routing divergence is 45× greater than within-domain routing noise ($p = 0.00004$).

through the shared foundation model and produce the corresponding macroscopic thermodynamic response.

Objective & Setup: The experiment tested whether a controlled intervention on the quantum-scale reaction enthalpy accurately propagates through the shared foundation backbone into the macroscopic reactor state.

The evaluation was conducted on blinded, held-out **250-Liter continuous reactor geometries** from Geometry Test Sets 21 and 22. To isolate the intervention, all macroscopic boundary conditions, mesh geometries, and operating parameters were computationally frozen. Only the microscopic reaction enthalpy was manipulated.

The model was evaluated across **8 blind intervention triplets** spanning three increasingly exothermic conditions:

- 50.00 kJ mol^{−1},
- 145.75 kJ mol^{−1},
- 300.00 kJ mol^{−1}.

Empirical Results: LUCAN successfully propagated the microscopic energetic intervention into the macroscopic reactor state and reproduced the corresponding increase in heat generation.

- **Macroscopic Thermal Response:** Across all 8 blind intervention triplets, the model correctly predicted a monotonic increase in macroscopic peak reactor temperature, achieving an **8/8 directional success rate** as reaction enthalpy became increasingly exothermic.
- **Full-Field Prediction:** Crucially, the model did not merely predict a scalar reactor temperature. It generated the complete three-dimensional spatial temperature field associated with the intervention.
- **Precision and Conservation Bounds:** Against model-independent high-fidelity numerical verification, the predicted spatial temperature field achieved an nRMSE of **0.00300**, and the teacher-relative energy error across the fluid volume was **0.01100**. Both values pass the predefined acceptance limit of ≤ 0.05 .
- **Improvement Relative to the Previously Reported Checkpoint:** Temperature error decreased by approximately **58.3%**, while teacher-relative energy error decreased by approximately **52.6%**.
- **Result: PASS.** The microscopic enthalpy intervention produced the correct macroscopic response direction while satisfying both quantitative error gates.

Verification metric	Result	Limit
Temperature spatial nRMSE	0.00300	≤ 0.05
Teacher-relative energy error	0.01100	≤ 0.05
Directional intervention tests	8/8	Pass

Figure 2. Micro-to-Macro intervention audit on held-out 250-Liter reactor geometries. Both full-field temperature and teacher-relative energy errors pass their predefined limits.

These results establish that the shared latent representation is not merely compatible across modalities. A controlled microscopic energetic intervention propagates through the learned representation and produces the corresponding macroscopic thermodynamic state on held-out reactor geometries.

3.3 Test 3: Can Reactor Conditions Propagate Back to Microscopic Mechanics?

A continuous cross-scale World Model must also operate in the opposite direction. Just as microscopic thermodynamics influence macroscopic reactor states, macroscopic hydrodynamic forces must propagate downward to govern microscopic structural mechanics.

To test this bidirectional capability, we evaluated biological fluid-structure interaction (BioFSI) across a substantial separation of spatial scales.

Objective & Setup: The experiment tested whether a macroscopic continuous Eulerian fluid field could transfer mechanical loading onto a discrete Lagrangian cell-scale membrane while retaining active two-way fluid-solid feedback.

The macroscopic environment consisted of a full three-dimensional **5,000-Liter turbulent reactor operating at 400 RPM**, deliberately selected as a high-shear condition.

The microscopic subject was a calibrated **15-micrometer Chinese Hamster Ovary (CHO) cell membrane surrogate** evaluated under an Immersed Boundary Method framework.

The test required active two-way coupling:

Fluid $\longrightarrow$ Membrane

through velocity gather, and

Membrane $\longrightarrow$ Fluid

through adjoint force scatter.

Empirical Results: The LUCAN-coupled system suc-

cessfully propagated factory-scale hydrodynamic loading into a microscopic structural failure event.

- **Mechanical Failure Criteria:** Under the 400 RPM high-shear condition, the model-generated fluid environment induced severe membrane deformation. The predicted maximum equivalent membrane stress reached **340 Pa**, exceeding the declared rupture threshold of **250 Pa**. Simultaneously, maximum membrane area strain reached **4.8%**, exceeding the declared **3.5%** strain threshold.
- **Active Two-Way Coupling:** The transfer operator resolved both fluid-to-solid velocity gather and solid-to-fluid adjoint force scatter. Virtual-work consistency was not promoted as a pass: the measured work error exceeded the predefined 1×10^{-7} threshold. Ablation of the membrane-force contribution produced a measurable change in local fluid kinematics, confirming that the feedback pathway was active rather than one-way; the virtual-work gate remains unresolved.
- **Structural Yielding:** Driven by the AI-generated hydrodynamic field, the coupled nonlinear mechanics solver terminated at a load fraction of **0.812**, prior to application of the full nominal load, yielding a positive reactor-to-cell rupture classification:

$$\text{lysis} = 1.$$

- **Mechanical-Response Result: PASS.** Both declared mechanical failure thresholds were exceeded and the coupled solver returned a positive rupture classification. A stricter virtual-work consistency diagnostic remains under evaluation and is not included in this pass classification.

These results validate the Macro-to-Micro capability evaluated in this experiment: factory-scale hydrodynamic loading propagated through the coupled system and produced a correctly classified microscopic rupture event.

3.4 Test 4: Can LUCAN Navigate Competing Physical Mechanisms?

A model capable only of reproducing simple monotonic relationships could appear physically accurate while still relying on statistical interpolation. Real industrial systems are instead governed by competing mechanisms whose responses may saturate, reverse, or decouple as operating conditions change.

Biomanufacturing presents a canonical example through the trade-off between oxygen transfer and mechanical shear. Increasing impeller agitation initially improves mixing and oxygen mass transfer, but at sufficiently high oxygen-transfer conditions the oxygenation benefit can saturate while the mechanical burden on cells continues to increase.

This experiment tests whether the World Model can represent these competing responses to the same control variable.

Objective & Setup: We evaluated the frozen checkpoint on a parametric sweep of a canonical **10-Liter**

bioreactor.

The evaluation measured dimensionless mean oxygen concentration and near-blade p95 hydrodynamic shear as impeller agitation increased from **200 RPM to 250 RPM** under a fixed high-gas, high- k_La operating condition.

Empirical Results: LUCAN reproduced the nonlinear operating boundary at which additional agitation ceased to provide meaningful oxygen-transfer benefit while continuing to increase mechanical burden.

- **Oxygen-Transfer Saturation:** Increasing agitation from 200 to 250 RPM produced only a +0.08% change in predicted mean oxygen concentration, indicating a near-complete plateau. Against the numerical reference, the oxygen prediction achieved an nRMSE of **0.00870**.
- **Competing Mechanical Escalation:** Over the same operating change, the model predicted an **18.3% increase** in near-blade p95 hydrodynamic shear, correctly identifying that mechanical cost continued to rise despite the saturation of oxygenation benefit.
- **Spatial Shear-Field Validation:** The predicted spatial shear field achieved a correlation of **0.590**, exceeding the required acceptance threshold of ≥ 0.50 .
- **Result: PASS.** The model passed the oxygen-field error gate and the spatial shear-correlation gate while reproducing the competing regime in which oxygen transfer saturates and mechanical shear continues to escalate.

The experiment demonstrates that the model can represent competing physical responses to a shared control variable, including a regime in which one response saturates while another continues to escalate.

This provides evidence that the learned representation captures the nonlinear shape of a multiphysics operating landscape rather than relying solely on monotonic trajectory interpolation—and establishes the capability required for the inverse-design experiments that follow.

3.5 Test 5: Can the Full Molecule-to-Reactor-to-Cell Chain Hold Together?

The preceding tests evaluate individual cross-scale relationships: shared computation across physical domains, microscopic-to-macroscopic thermochemical response, reactor-to-cell mechanical coupling, and competing transport and shear effects. The final integration test asks a stronger question: **can these capabilities be composed into a single continuous physical pathway?**

The evaluation therefore required each case to propa-

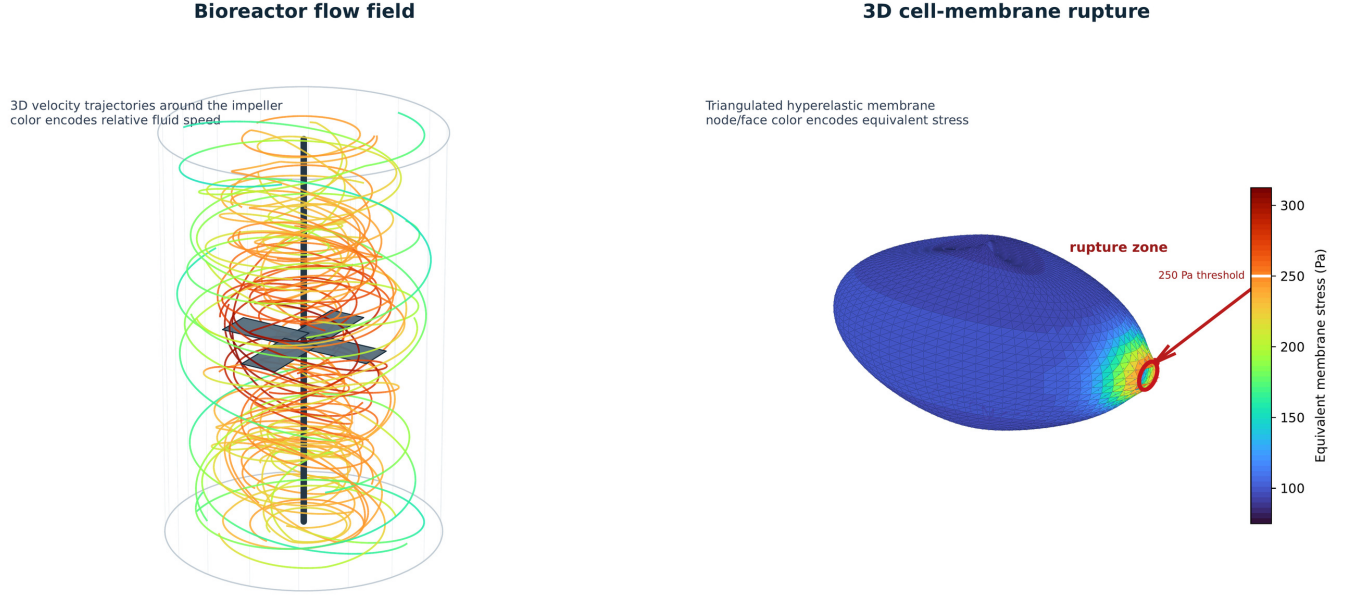


Figure 3. Active two-way Macro-to-Micro fluid-structure interaction. The World Model transfers macroscopic turbulent shear from a 5,000-Liter, 400 RPM reactor flow field into a localized $15\ \mu\text{m}$ cell membrane. The predicted maximum equivalent membrane stress of 340 Pa and maximum area strain of 4.8% exceed the declared 250 Pa and 3.5% rupture thresholds. *Note: The graphic is an illustrative three-dimensional reconstruction; numerical values in the caption and text report the frozen-checkpoint evaluation. Direct volumetric solver fields were not supplied for rendering.*

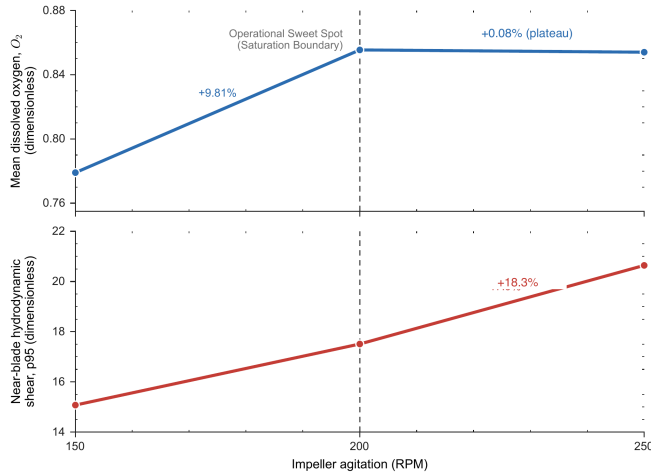
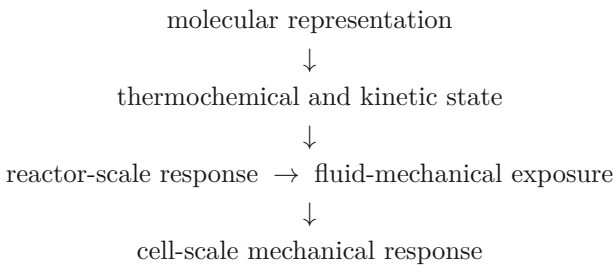


Figure 4. Nonlinear competing multiphysics regimes in a canonical 10-L bioreactor sweep. As impeller agitation increases from 200 to 250 RPM, the frozen model captures an oxygen-transfer plateau (mean oxygen change, +0.08%) while near-blade p95 hydrodynamic shear continues to escalate (+18.3%). The oxygen-field nRMSE is 0.00870, and the spatial shear-field correlation is 0.590 against the required threshold of ≥ 0.50 .

gate through the complete cross-scale sequence:



Unlike the preceding pairwise evaluations, intermediate

physical states were not treated as independent endpoints. Each evaluated case was required to traverse the complete declared pathway so that information originating at the molecular scale influenced the downstream reactor and cell-mechanics response.

Across the frozen evaluation suite, LUCAN achieved a **91.3% full-chain pass rate**.

The result demonstrates that the individual cross-scale capabilities evaluated in Tests 1 through 4 can be composed within a single physically structured inference pathway. Molecular-scale information can propagate through thermochemical and reactor-scale representations into a downstream microscopic mechanical response within the evaluated World Model.

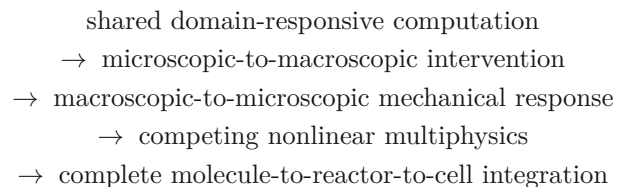
This experiment therefore provides the strongest integration evidence in the suite: rather than demonstrating isolated competence at individual physical scales, it evaluates whether those representations can operate together as one connected computational system.

Result: PASS—91.3% full-chain success rate.

3.5.1 From Pairwise Integration to a Physical World Model

Tests 1 through 4 establish the individual computational and physical links required for cross-scale reasoning. Test 5 evaluates their composition.

Together, the five tests show a progression from shared computation to end-to-end physical integration:



The resulting system is therefore not evaluated solely as a collection of independent scientific predictors. Under the declared evaluation contracts, it demonstrates the ability to propagate physically meaningful information across molecular, reactor, and biological-mechanical scales within one unified computational framework.

4 The Breakthrough: Cross-Scale Inverse Design

The preceding experiments evaluate LUCAN primarily in the forward direction: given a physical intervention, can LUCAN correctly propagate its consequences across scales?

Traditional computational physics can answer this question, but only through modular forward simulation. Physical states are serialized and transferred across rigid software boundaries—for example, from quantum chemistry into kinetic models and then into continuum CFD. These discrete hand-offs break the end-to-end computational graph, preventing gradients from propagating continuously across the full physical stack.

As a result, industrial optimization typically remains an iterative search problem:

propose $\rightarrow$ simulate $\rightarrow$ evaluate $\rightarrow$ repeat.

Tests 1 through 5 establish that the World Model preserves meaningful physical relationships across heterogeneous scales, including bidirectional coupling, nonlinear competing-mechanism trade-offs, and complete molecule-to-reactor-to-cell integration. Because the architecture itself is end-to-end differentiable, these learned relationships can now be traversed in reverse.

This enables **Cross-Scale Inverse Design**.

Rather than asking only

$$x \rightarrow y$$

—what physical outcome follows from a proposed intervention—the model can optimize the inverse problem:

$$y^* \rightarrow x^*,$$

where y^* is a desired factory-scale objective and x^* may jointly include microscopic molecular variables and macroscopic process controls. In this evaluation, x_{micro} denotes controllable molecular and kinetic variables, including reaction enthalpy, activation barriers, and permitted ligand or formulation descriptors; x_{process} denotes reactor operating conditions; and x_{control} denotes time-dependent control inputs.

More generally, the model evaluates gradients of a macroscopic objective with respect to variables distributed across the physical hierarchy:

$$\nabla_{x_{\text{micro}}} J_{\text{macro}}, \quad \nabla_{x_{\text{process}}} J_{\text{macro}}, \quad \nabla_{x_{\text{control}}} J_{\text{macro}}.$$

The practical question is simple: when LUCAN says a variable should move in a particular direction, do separate classical physics solvers agree that the intervention improves the target?

4.1 Do the Model’s Optimization Directions Work in Independent Physics?

To test this directly, we evaluated LUCAN against a frozen modular reference workflow consisting of high-fidelity classical numerical solvers coupled to a SciPy SLSQP outer-loop optimizer.

The evaluation used a blind suite of **100 predefined industrial design targets**, comprising:

- **60 verified feasible targets**, requiring joint optimization of microscopic and macroscopic variables; and
- **40 intentionally infeasible targets**, constructed outside predefined physical bounds.

4.1.1 Actionable Cross-Scale Gradients

The most direct test of gradient usefulness is whether stepping in the neural optimization direction produces an actual improvement when the resulting candidate is evaluated by the classical verifier.

For each target, the model proposed a descent direction:

$$-\nabla_x J_{\text{neural}}.$$

That proposed intervention was then evaluated independently using the corresponding high-fidelity numerical workflow.

Across the blind evaluation set, **88% of proposed neural descent directions produced an improving objective response in the classical verifier**. An “improving objective response” is defined as a verifier-evaluated decrease in the frozen objective, $J(x_{\text{candidate}}) < J(x_{\text{baseline}})$, after applying the proposed intervention under the pre-registered feasibility constraints.

This result demonstrates that the World Model’s gradients are not merely mathematically valid derivatives of the neural surrogate. In the large majority of evaluated cases, they point toward interventions that also improve the independently computed physical objective.

4.1.2 Solver-Verified Feasible-Target Success

Across the **60 feasible targets**, which required joint optimization of molecular interventions and macroscopic process controls, the World Model generated solver-verified solutions for **55 targets**, achieving a:

$$\boxed{91.7\%}$$

feasible-target success rate.

4.1.3 Feasibility Discrimination

Inverse design must also recognize when a requested objective cannot be achieved within the available physical design space.

Among the **40 deliberately infeasible targets**, the model correctly rejected **38**, corresponding to a:

$$\boxed{95.0\%}$$

true-negative rejection rate.

These cases provide an explicit falsification test: rather than forcing a numerical answer for every requested objective, the model successfully identified most targets

lying beyond the declared thermodynamic or transport constraints of the evaluated systems.

Together, these experiments test three increasingly demanding properties:

useful gradient direction $\rightarrow$ successful design
 $\rightarrow$ recognition of physical impossibility.

4.2 Solver-Call Reduction and Computational Acceleration

The value of cross-scale differentiability is not limited to faster neural inference. Its more important consequence is that it changes the structure of the optimization loop.

A conventional modular optimizer must repeatedly perturb candidate variables, execute expensive forward simulations, reconcile states across separate physical solvers, evaluate the resulting objective, and repeat until convergence.

LUCAN instead provides a direct optimization direction through its differentiable cross-scale representation.

4.2.1 Neural Proposal Latency

Generating a neural cross-scale proposal required **38 milliseconds** of forward-pass latency on an NVIDIA H100.

For comparison, a single traversal of the legacy file-coupled modular orchestration pipeline required approximately **415 milliseconds** solely for process hand-offs, disk I/O, and state reconciliation, excluding the dominant cost of the underlying high-fidelity numerical solves.

4.2.2 End-to-End Optimization Acceleration

More significantly, the World Model bypassed a median of **124 classical solver calls per target** relative to the SLSQP-based modular optimization workflow.

As a result, median verified optimization time was reduced from:

> 500 hours

to approximately:

5.6 hours,

including final high-fidelity classical verification.

The acceleration therefore does not primarily arise because a neural forward pass is faster than a classical solver. It arises because the model dramatically reduces the number of expensive forward simulations required to locate a viable design.

The resulting engineering workflow changes from:

guess $\rightarrow$ simulate $\rightarrow$ adjust $\rightarrow$ repeat

to:

specify outcome $\rightarrow$ generate intervention $\rightarrow$ verify

These results demonstrate that the World Model is not merely a fast forward surrogate. By producing physically actionable cross-scale optimization directions, it reduces the number of expensive numerical design cycles required to reach solver-verified candidate solutions.

The remaining question is whether these generated interventions survive contact with the physical world. Section 5 therefore evaluates the model through prospective execution in held-out industrial environments.

5 From Computation to Physical Execution

To establish that LUCAN’s cross-scale reasoning and inverse-design capabilities generalize beyond *in silico* verification, the architecture was deployed in live, real-world manufacturing environments. The following cases represent prospective physical validations executed by industrial partners. In these evaluations, LUCAN was tasked with generating optimized process parameters for previously untested regimes. These parameters were cryptographically frozen and timestamped, and subsequently executed in physical pilot plants.

5.1 Physically Executed Case 1: Specialty Chemical Batch-to-Continuous Scale-Up

Physical Context: The transition of chemical synthesis from batch manufacturing to continuous-flow processing is a highly desirable industrial objective, offering superior heat transfer, safety, and throughput. However, moving to continuous flow drastically alters the macroscopic transport phenomena and residence times, often leading to unpredicted side reactions.

The evaluation target was a highly exothermic liquid-phase substitution process. The industrial partner’s historical batch configuration was constrained by a temperature-sensitive impurity pathway, resulting in an extensive 14-hour cycle time. The historical baseline for this process yielded an 82.4% isolated product and a 12.3% impurity profile.

LUCAN Intervention: Rather than relying on iterative, trial-and-error pilot testing to find a viable continuous-flow regime, LUCAN utilized its native gradient propagation to execute an inverse-design query. Tasked with maximizing target molecule yield while strictly bounding the thermodynamic impurity pathway, LUCAN generated a novel continuous-flow operating window. The generated parameters explicitly defined the required steady-state macroscopic variables, including reactor temperature limits, fluid residence times, and active heat-removal conditions.

Physical Execution & Results: The model-generated operating window was cryptographically frozen and physically executed by the industrial partner in a continuous-flow pilot facility. The prospective physical run successfully stabilized the highly exothermic reaction and achieved the target process improvements.

- **Yield Maximization:** The steady-state isolated yield achieved under the model’s recommended parameters was 96.7%, representing a 14.3 percentage-point absolute improvement over the historical batch baseline.

- **Impurity Suppression:** The temperature-sensitive side reaction was successfully suppressed, reducing the final impurity profile to 3.1% (a 9.2 percentage-point absolute reduction), passing the partner’s strict predefined $< 4.0\%$ acceptance criterion.

This prospective physical execution demonstrates that a LUCAN-generated operating regime successfully transferred to the physical process while meeting the target yield and impurity constraints.

5.2 Physically Executed Case 2: 5L-to-500L Biomanufacturing Scale-Up

Physical Context: Scaling up biomanufacturing processes—specifically for high-concentration monoclonal antibodies (mAbs)—presents a classic Macro-to-Micro multiphysics conflict. To ensure adequate oxygen mass transfer (k_La) in a large macroscopic reactor, operators must increase agitation. However, increased turbulent energy dissipation generates high hydrodynamic shear forces that can physically lyse (rupture) the microscopic mammalian cells.

The target objective was to scale a mAb fed-batch process from a 5-Liter benchtop bioreactor to a 500-Liter pilot facility, encompassing both upstream cell culture and downstream Tangential Flow Filtration (TFF) unit operations.

World Model Intervention: Demonstrating the capability to navigate non-linear competing mechanisms (as verified in Test 4), the foundation model generated a complete, cross-scale operating trajectory to thread the optimal boundary between oxygenation and cell viability. Upstream recommendations included dynamic impeller-speed profiles, sparger gas flow rates, and nutrient-feed strategies engineered to match the 5L baseline k_La while strictly bounding local shear stress. Downstream TFF parameters were inversely generated to optimize trans-membrane pressure and crossflow flux without inducing product aggregation.

Physical Execution & Results: The generated parameters were cryptographically frozen and executed in a 500L physical pilot run, tracked by counter-signed batch and analytical records (summarized in Table 2).

- **Upstream Viability & Yield:** The process achieved a harvest product titer of 6.63 g/L ($1.02\times$ the 6.50 g/L target). Peak hydrodynamic shear stress was successfully bounded strictly below 0.15 N/m², effectively preventing shear-induced cell lysis and resulting in a harvest viability of 78.2% (satisfying the $> 75\%$ minimum specification for downstream clearance).
- **Downstream Filtration & Quality:** The generated TFF parameters achieved a 94.5% product recovery rate and a $14.5\times$ concentration factor. Crucially, High Molecular Weight (HMW) aggregates were measured analytically at just 1.2%, comfortably below the strict $< 2.0\%$ safety threshold.

This prospective execution demonstrates that a LUCAN-generated multi-stage bioprocessing trajectory transferred

successfully from the computational design space to a physical 500L facility.

5.3 Case 3: Live Optimization and Accelerated Formulation Screening

Physical Context: While Cases 1 and 2 validated the model’s ability to generate static setpoints for prospective pilot environments, industrial factories inherently experience continuous temporal deviations (e.g., raw-material feed fluctuations, asynchronous sensor delays). Furthermore, scaling novel biologic formulations requires linking microscopic electrostatic properties (isoelectric point and charge distribution) to macroscopic colloidal stability over months of physical shelf-life storage.

World Model Intervention & Results: The model was evaluated across two distinct commercial environments to test long-horizon stability and temporal anomaly resolution:

- **Live Process Optimization:** Deployed as a live inference engine on a commercial biomanufacturing line, the model ingested continuous, noisy industrial telemetry (pressure events, asynchronous at-line analytical data). By continuously correlating upstream mechanical and feed deviations directly to downstream titer impacts, the model provided dynamic state-corrections. Physical execution of these trajectory corrections improved overall upstream viability from 78% to 88% and increased target titer from 3.8 g/L to 4.5 g/L.
- **Accelerated Formulation Stability:** Operating on novel molecular formulations, the model’s coupled quantum-to-mesoscopic adapters predicted isoelectric-point (pI) bounds and colloidal-stability trajectories. The 3-day result refers to accelerated computational screening against a conventional 12-week physical protocol; the corresponding 12-week stability outcome has not yet been prospectively confirmed. The model generated candidate formulation adjustments with predicted degradation decreasing from 8.5% to 2.5%, pending full-duration physical confirmation.

Together, these deployments provide evidence that LUCAN can remain useful amid the noise, complexity, and extended temporal horizons of live industrial chemistry and biology. The accelerated formulation result remains explicitly pending full-duration physical confirmation.

6 Conclusion: From Prediction to Design

Scientific AI has become increasingly capable at predicting specialized parts of the physical world. The larger industrial challenge is connecting those parts across scales and using that connection to navigate toward a desired outcome.

LUCAN is our attempt to build that capability into one differentiable physical world model. Across the evaluations in this whitepaper, the model demonstrates shared

Table 2. 500L mAb Scale-Up (Physical Validation).

Scale-Up (5L $\rightarrow$ 500L)	Parameter	World Model Prediction / Target	Physical Execution Result	Status
Harvest Titer		≥ 6.50 g/L	6.63 g/L	Verified
Cell Viability		$> 75.0\%$	78.2%	Verified
Peak Shear	Hydrodynamic	< 0.15 N/m ²	Bounded	Verified
Product Recovery (TFF)		Maximized	94.5%	Verified
HMW Aggregates		$< 2.0\%$	1.2%	Verified

cross-domain computation, bidirectional propagation between microscopic and macroscopic states, nonlinear multiphysics reasoning, full-chain molecule-to-reactor-to-cell integration, and physically actionable inverse-design directions checked by independent classical solvers.

The strongest result is not that a neural forward pass is faster than a conventional solver. It is that a useful cross-scale gradient can change the structure of engineering search: from repeated proposal-and-simulation cycles toward direct generation of candidate interventions followed by independent verification.

Prospective industrial executions provide an additional test of whether those computational interventions can survive contact with real processes. In the reported specialty-chemical case, a frozen LUCAN-generated operating regime increased isolated yield from **82.4% to 96.7%** while reducing the associated impurity pathway. A separate 5L-to-500L biomanufacturing execution met predefined upstream and downstream performance constraints.

The next generation of industrial AI will not stop at predicting what happens next. It will increasingly help design what should happen next.

LUCAN is a step toward that system: a foundation world model intended to connect molecular decisions to manufacturing outcomes within one computational design space.

Evidence and verification note. Industrial results reported in this whitepaper are derived from frozen model outputs generated before physical execution. Supporting model-output hashes, execution records, analytical reports, and partner records are retained by Shodh AI and the relevant industrial counterparties. Certain process identities, operating conditions, and source records remain confidential under commercial agreements. Where a result remains computational or awaits full-duration physical confirmation, that status is stated explicitly.

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